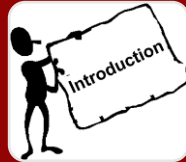


On your tutorials



UNIVERSIDAD
DE MÁLAGA





What is flamapy?



flamapy as a tool



flamapy as a framework

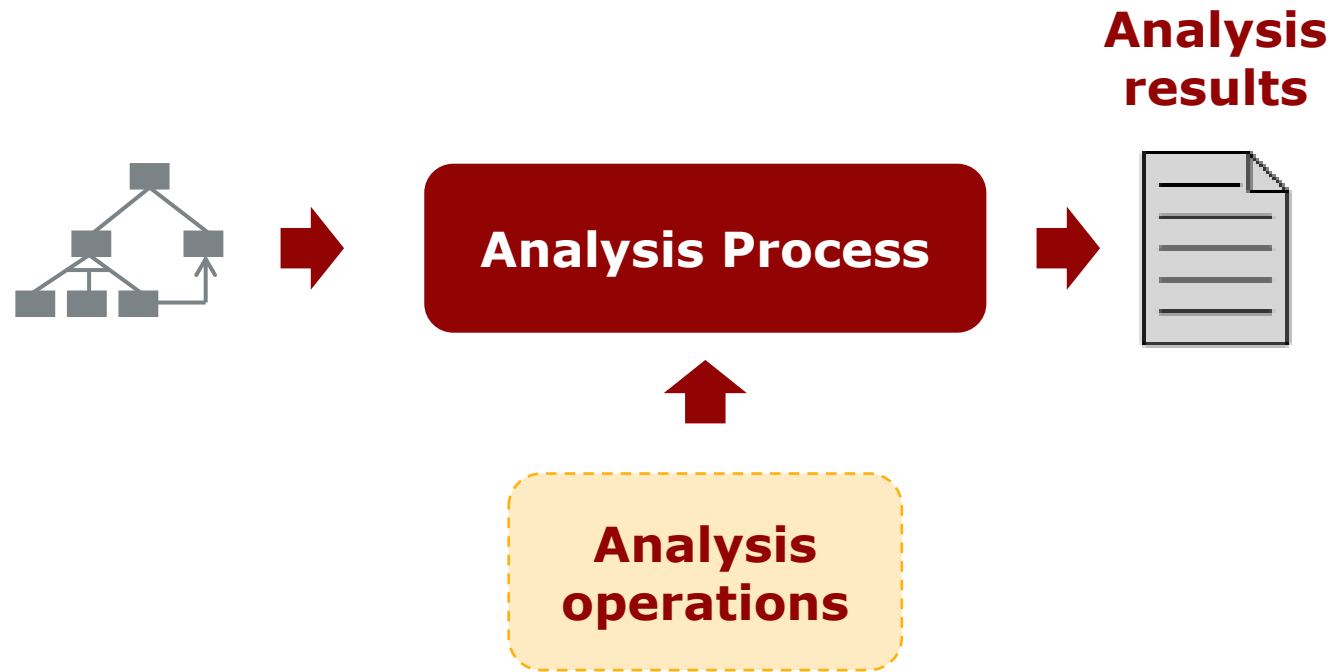


Let's get dirty

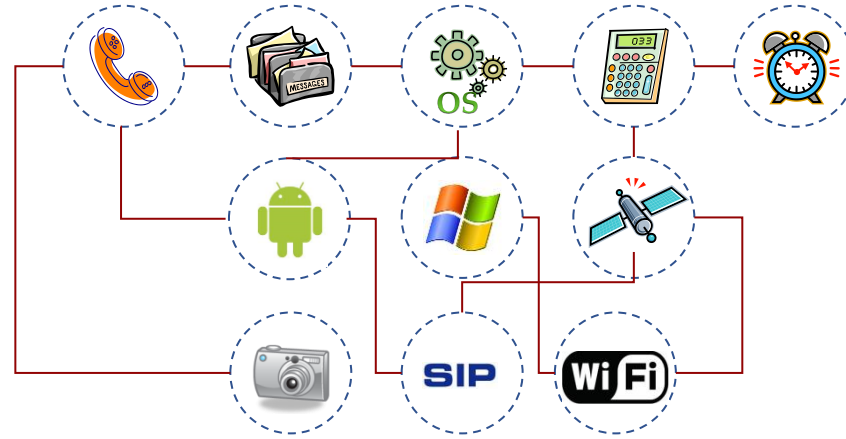
Variability Models



Automated Analysis



Analysis Operations



**My app requires
wifi and bluetooth 4**



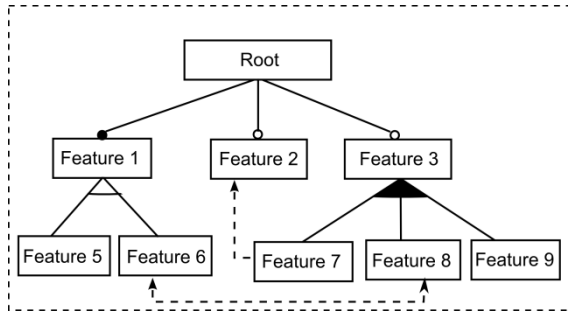
**In how many different
devices will it work?**



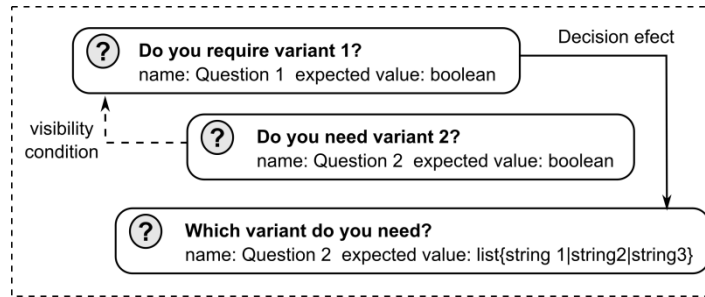
A tool and a framework for the

Automated Analysis Of Variability Models

Variability models... but feature modelling

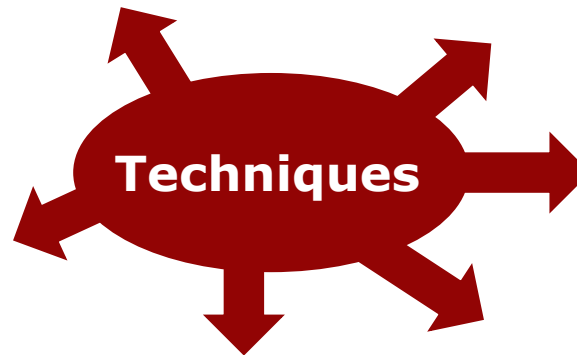


Feature modelling

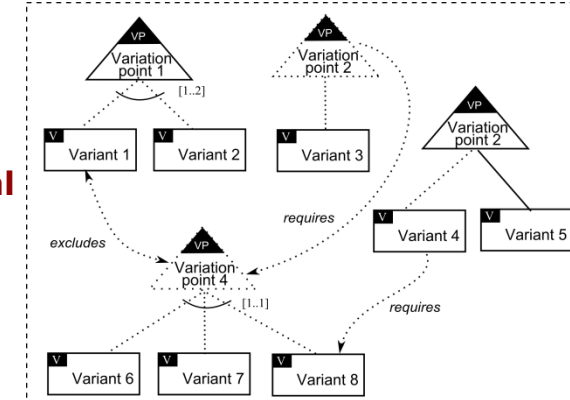


Decision modelling

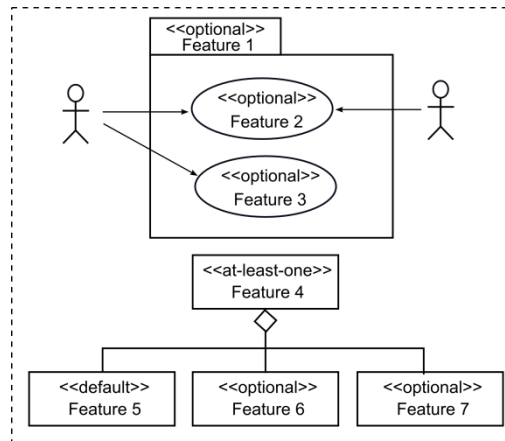
Ad-hoc solutions:
tables, textual
docs, ...



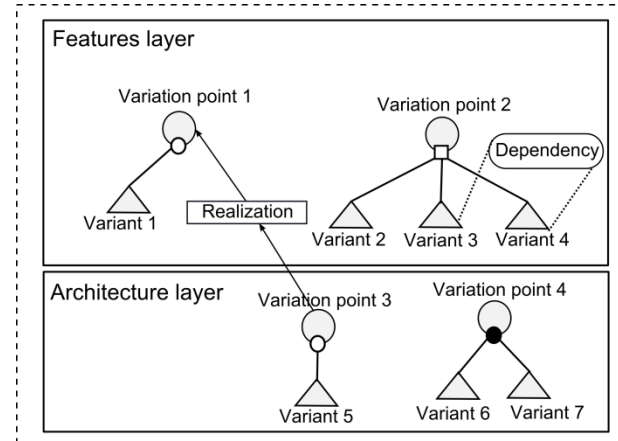
Orthogonal
variability
modelling



UML-based



COVAMOF



A tool and a framework

As a tool

Allows to analyze feature models

Multiple interfaces

- CMD
- Python
- WASM

As a framework

Adapt it to what you need

- Allow other variability models
- Create your:
 - Operations
 - Transformations
 - Solvers

Integrate it in your tools



What is flamapy?



flamapy as a tool



flamapy as a framework



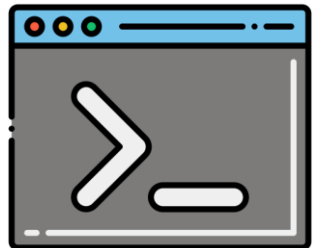
Let's get dirty

Flama as a tool

- Simple installation
 - `pip install flamapy`
 - Installs the underlying framework
 - The pysat and bdd solvers (both well tested with Betty tool suite)
 - The easy-to-use Python facade

```
!pip install flamapy

Collecting flamapy
  Downloading flamapy-2.0.1-py3-none-any.whl.metadata (7.9 kB)
Requirement already satisfied: wheel in /usr/local/lib/python3.10/dist-packages (from flamapy) (0.44.0)
Collecting flamapy-fw~2.0.1 (from flamapy)
  Downloading flamapy_fw-2.0.1-py3-none-any.whl.metadata (2.7 kB)
Collecting flamapy-fm~2.0.0 (from flamapy)
  Downloading flamapy_fm-2.0.1-py3-none-any.whl.metadata (963 bytes)
Collecting flamapy-sat~2.0.1 (from flamapy)
  Downloading flamapy_sat-2.0.1-py3-none-any.whl.metadata (1.1 kB)
Collecting flamapy-bdd~2.0.1 (from flamapy)
  Downloading flamapy_bdd-2.0.1-py3-none-any.whl.metadata (8.2 kB)
```



Supported formats

- UVL
- Fama XML
- Feature IDE
- Glencoe
- Others (CNF, JSON, ...)

```
include
  Boolean.group-cardinality
  Arithmetic.aggregate-function
  Arithmetic.feature-cardinality
  Type

features
  Sandwich
    mandatory
      Bread {Calories 100, Sugar 20}
    optional
      Sauce
        or
          Ketchup {Calories 40, Sugar 35}
          Mustard {Calories 25, Sugar 5}
      Cheese
        [0..2] // Group cardinality
        Cheddar {Calories 60}
        Gouda {Calories 50}
        Goat {Calories 35}
      Pickle cardinality [1..3] // Feature cardinality

constraints
  Ketchup => Cheese
  Bread.Sugar + Ketchup.Sugar + Mustard.Sugar < 60 // Attribute constraints
  sum(Calories) < 160 // Attribute aggregate
```

Analysis operations

Atomic Sets	Average Branching Factor	Commonality	Configurations
Number of Configurations	Conflict Detection	Core Features	Count Leafs
Dead Features	Estimated Number of Configurations	False Optional Features	Feature Ancestors
Filter	Leaf Features	Max Depth	Satisfiable
	Satisfiable Configuration	Unique Features	

SpringerBriefs in Computer Science

Alexander Felfernig · Andreas Falkner ·
David Benavides



Feature Models

AI-Driven Design, Analysis
and Applications

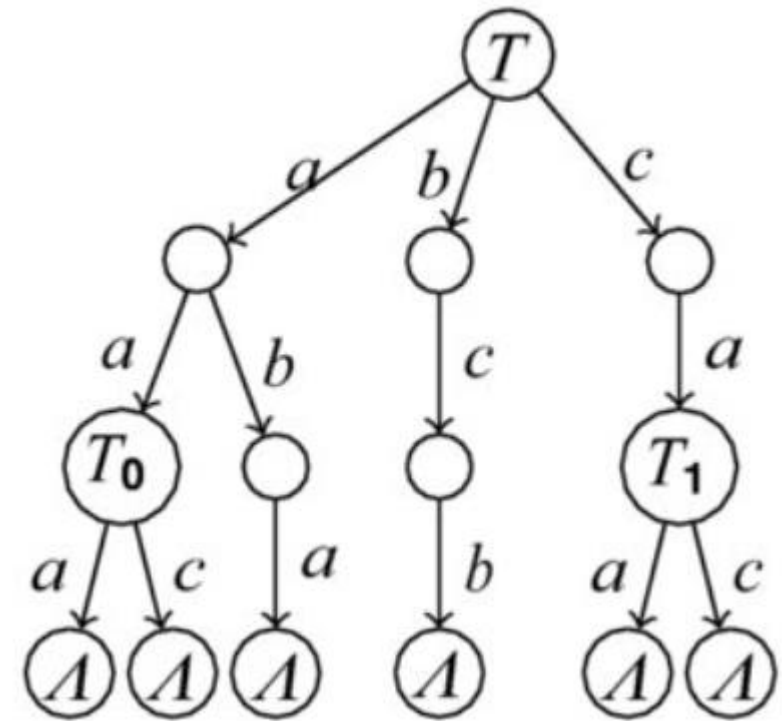
OPEN ACCESS

 Springer

Solvers supported

PySAT 

CaDiCaL (rel-1.0.3)
CaDiCaL (rel-1.5.3)
CaDiCaL (rel-1.9.5)
Glucose (3.0)
Glucose (4.1)
Glucose (4.2.1)
Lingeling (bbc-9230380-160707)
MapleLCMDistChronoBT (SAT competition 2018 version)
MapleCM (SAT competition 2018 version)
Maplesat (MapleCOMSPS_LRB)
Mergesat (3.0)
Minicard (1.2)
Minisat (2.2 release)
Minisat (GitHub version)



CUDD Solver

Interfaces

- WASM
 - <https://www.flamapy.org/try-it/wasm-uvl-analysis>
- Python (FAMILIAR alike)
 - https://docs.flamapy.org/tool/python_facade/
- CMD
 - https://docs.flamapy.org/tool/command_line_interface

Let's play with it



<https://www.flamapy.org/try-it>



What is flamapy?



flamapy as a tool



flamapy as a framework



Let's get dirty

Contact

FaMa Tool Suite

ISA

Search... Location: Welcome

Welcome

FaMa-FW is a Framework for automated analyses of feature models integrating some of the most commonly used logic representations and solvers proposed in the literature (BDD, SAT and CSP solvers are implemented). FaMa is the first tool integrating different solvers for the automated analyses of feature models.

FaMa FW
Download

Get It!

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Template by
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Last update: April 08, 2019
15:17:23

UR

If you are interested in this tool, please feel free to [make a request](#).
For more information, contact us at fama.support@gmail.com

News

- [FaMa 1.1.2 Released](#)
March 08, 2012
- [FaMa 1.1.1-2 Released](#)
April 14, 2011
- [FaMa 1.1.1 Released](#)
March 31, 2011
- [FaMa 1.1.0 Released](#)
January 30, 2011
- [FaMa 1.0.0 Final](#)
July 21, 2010

RSS

Sitemap
Print Version

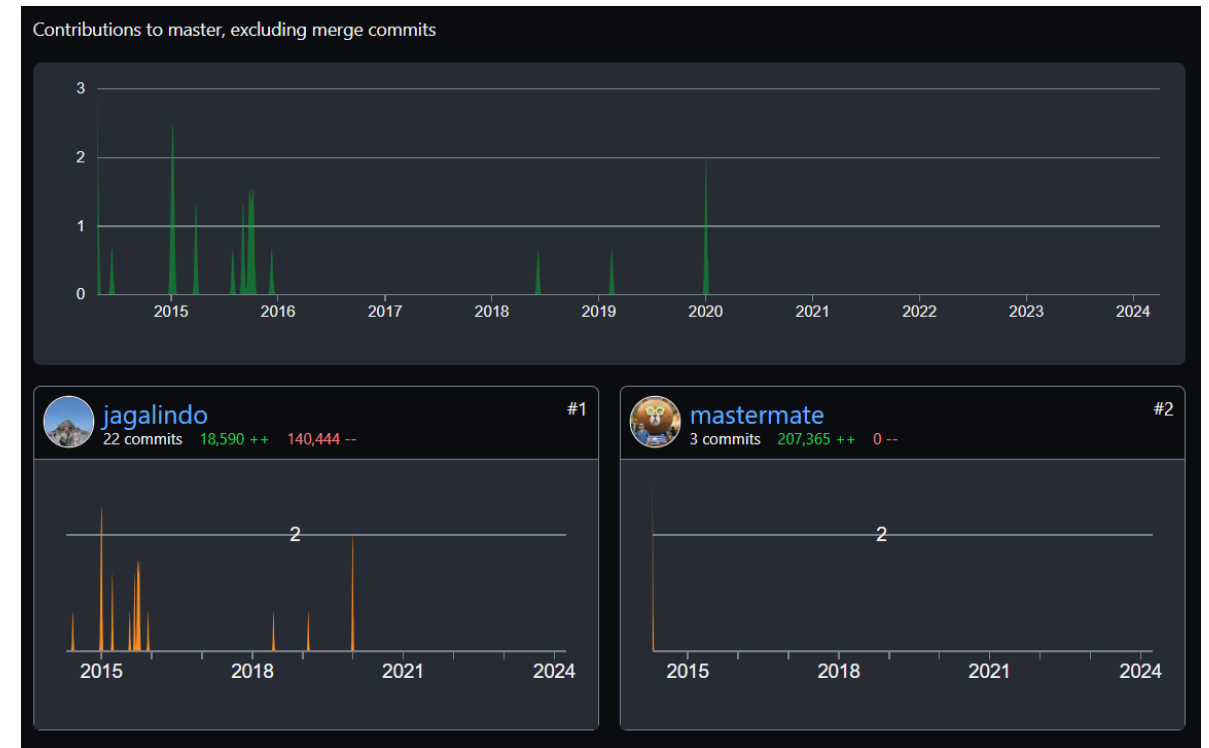
FaMa Test Suite 1.2

FAMA FW is free software; you can redistribute it and/or modify it under the terms of the [GNU Lesser General Public License](#).

OSGi COMPLIANT

LFPIV3

This work was partially supported by the European Commission (FEDER) and Spanish Government under Web-Factories (TIN2006-00472) and SETI (TIN2009-07366) projects and by the Andalusian Government under project ISABEL (TIC-2533).



Our past experience

Identifying problems in three variability analysis community

What drive flamapy framework design?

Problems



Researchers
EGO

I won't contribute to others tool

Please just PR me

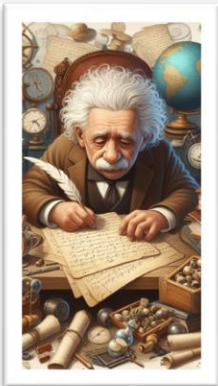
It's my tool, then my rules



Researchers'
niche

Don't look to what others are doing

No integration between tools



Technical debt

Java 6 is the newest ecosystem we do find

Most projects focus on adding new functionality
from their devs/researchers.

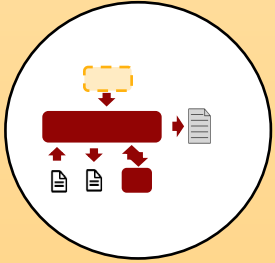
Tests are for riches'

Difficult to execute

A plugin based framework?

pip install flamapy-fw

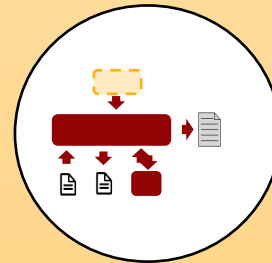
Core



- Operations interfaces
- Discover and orchestrator
- Common concepts

pip install flamapy-fm

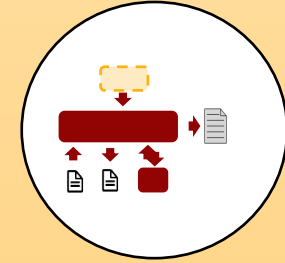
FM



- Metaclasses for FM
- Operations for FM
- Transformation to other metamodels

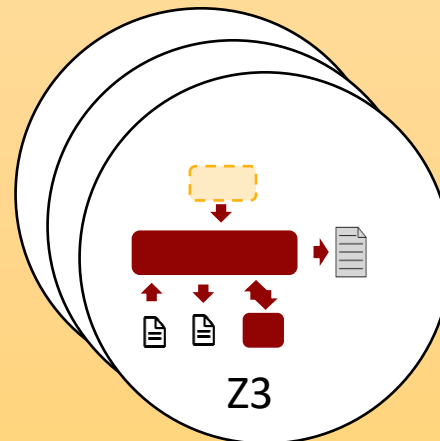
pip install flamapy-sat

SAT



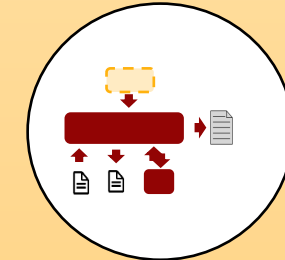
- Metaclasses for SAT
- Operations for SAT
- Transformation to other metamodels

- Each researcher can have their own plugin with their own rules
- There is a basic “distribution” that just installs the basic plugins, but others can be integrated



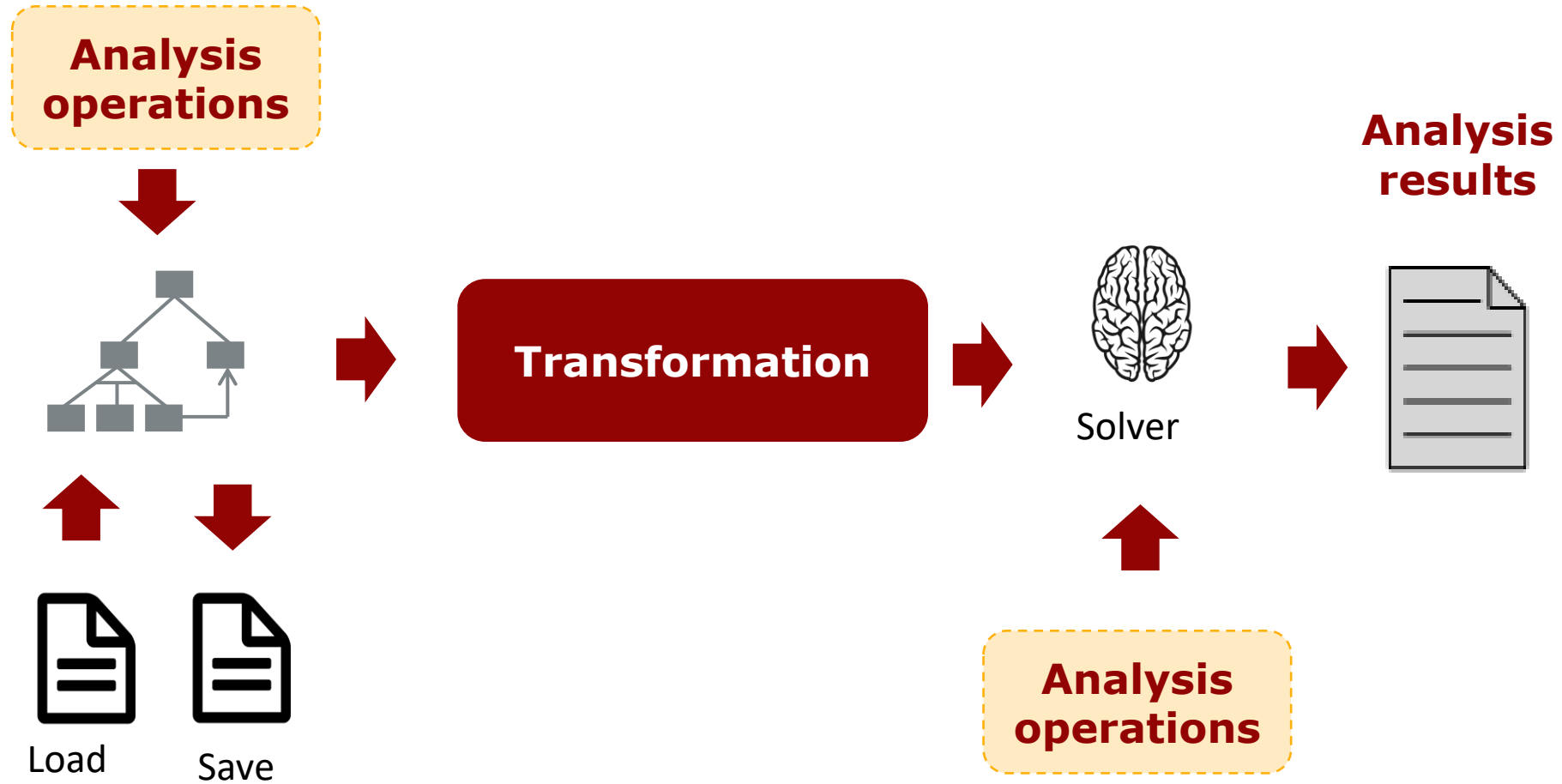
pip install flamapy-fw

BDD

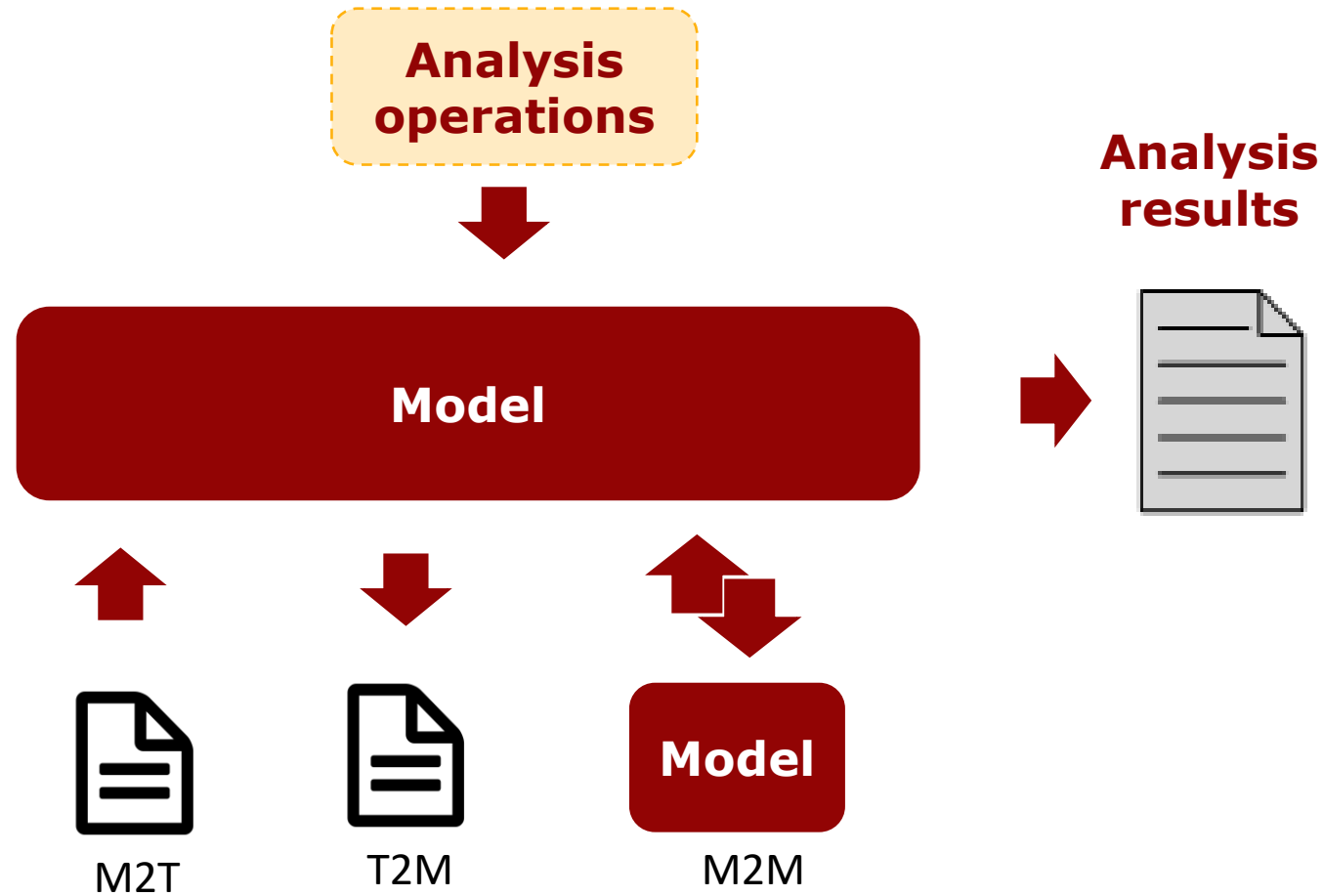


- Metaclasses for BDD
- Operations for BDD
- Transformation to other metamodels

The old metamodeling approach



The “new” metamodeling approach

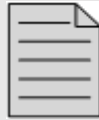


FM metamodel

Count leafs
Count products
approximated
...

FM metamodel

**Analysis
results**



- Fide writer
- UVL writer
- ...



- UVL reader
- FamaXML reader
- SPLX reader
- ...

**m2
m**

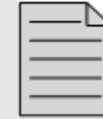
- ...

Pysat metamodel

#Products
Valid configuration
Valid Product
...

**PySAT
metamodel**

**Analysis
results**



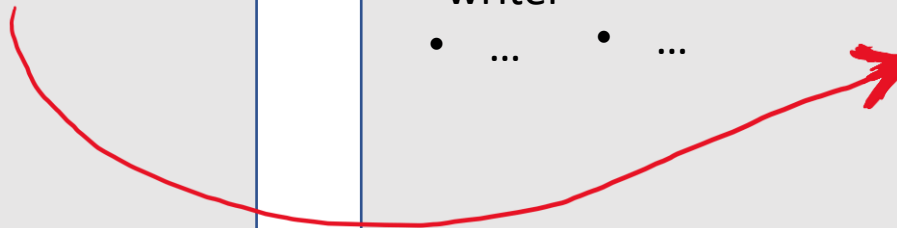
- CNF writer
- ...



- CNF reader
- ...

**m2
m**

- Feature model to SAT



Let's see how it works

Configurations

Basic usage

```
from flamapy.core.discover import DiscoverMetamodels
dm = DiscoverMetamodels()
result = dm.use_operation_from_file("PySATConfigurations", "./samplemodel.uv1")
print(result)
```

[Configuration({'eCommerce': True, 'Server': True, 'PHP': True, 'v74': True, 'Storage

Advanced usage

```
[ ] from flamapy.core.discover import DiscoverMetamodels
dm = DiscoverMetamodels()
feature_model = dm.use_transformation_t2m("./samplemodel.uv1", 'fm')
sat_model = dm.use_transformation_m2m(feature_model, "pysat")
operation = dm.get_operation(sat_model, 'PySATConfigurations')
operation.execute(sat_model)
result = operation.get_result()
print(result)
```

[Configuration({'eCommerce': True, 'Server': True, 'PHP': True, 'v74': True, 'Storage



<https://www.flamapy.org/try-it>

Integrating it with other tools

- Using the framework enables easy integration with other tools.
 - Fact label was the first integration of flamapy within another tool
 - <https://fmfactlabel.adabyron.uma.es/>



FM fact label: a configurable and interactive visualization of feature model characterizations

Authors:  [Jose M. Horcas](#),  [Jose A. Galindo](#),  [Mónica Pinto](#),  [Lidia Fuentes](#),  [David Benavides](#) | [Authors Info & Claims](#)

SPLC '22: Proceedings of the 26th ACM International Systems and Software Product Line Conference - Volume B • Pages 42 - 45

<https://doi.org/10.1145/3503229.3547025>

Linux 2.6.33.3 simple		
Version of the Linux kernel FM in which pseudo-complex constraints have been converted into simple (requires and excludes) constraints.		
Tags: Linux, KConfig, OS, real-world FM		
Author: Alexander Knüppel et al.		
Year: 2017		
Domain: Operating systems		
Features	6467	
Abstract features	0	(0%)
Abstract leaf features	0	(0%)
Abstract compound features	0	(0%)
Concrete features	6467	(100%)
Concrete leaf features	5523	(85%)
Concrete compound features	944	(15%)
Compound features	944	(15%)
Leaf features	5523	(85%)
Root feature	1	(0%)
Top features	1004	(16%)
Solitary features	6281	(97%)
Grouped features	185	(3%)
Tree relationships	6322	
Mandatory features	244	(4%)
Optional features	6037	(96%)
Feature groups	41	(1%)
Alternative groups	39	(95%)
Or groups	2	(5%)
Mutex groups	0	(0%)
Cardinality groups	0	(0%)
Depth of tree	7	
Max depth of tree	7	
Mean depth of tree	2.7	
Median depth of tree	3	
Branching factor	6.85	
Avg children per feature	1	
Min children per feature	1	
Max children per feature	1004	
Cross-tree constraints	7650	
Simple constraints	7313	(96%)
Requires constraints	7108	(97%)
Excludes constraints	205	(3%)
Complex constraints	337	(4%)
Pseudo-complex constraints	0	(0%)
Strict-complex constraints	337	(100%)
Features in constraints	2943	(46%)
Avg constraints per feature	2.38	
Min constraints per feature	0	
Max constraints per feature	517	
Valid (not void)	Yes	
Core features	53	(1%)
Dead features	211	(3%)
Variant features	6203	(96%)
False-optional features	39	(1%)
Configurations	≤ 3.90e1672	

How to integrate it within UVL-hub



flamapy



University of Seville

The university of Seville is currently providing support for the core and the pysat plugins.



University of Málaga

The university of Malaga is currently providing support for the core and the bdd plugins.



University of Graz

The university of Graz is currently providing support for the operations regarding the explanation of errors in feature modeling.



Universidad Nacional de Educación a Distancia

The UNED is currently providing support for the implementation of BDD techniques that cope with colossal feature models.

